

# Lab 1 Report

Character-Level Language Model Pretraining, Sentiment Classification, and CycleGAN Image Style Transfer

**Team 09: PairProgramming\_Team\_09**

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Shared repository: <https://github.com/Shibin-1020/Gen-AI---LAB-1>

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# 1. Team ownership and reproducibility

Shibin Thomas and Denisha Ketan Tank each independently implemented and trained a model for every task in their own member folders. Shibin contributed the shared repository structure, shared evaluation utilities, smoke tests, and his independent Task 1, Task 2, and Task 3 runs. Denisha contributed an independent from-scratch character GPT, three sentiment classifiers, an independent CycleGAN run, official Task 3 evaluation, generated outputs, and a Kaggle submission. The comparison and report were assembled as a team.

Repository layout follows task1\_llm/, task2\_sentiment/, and task3\_gan/. Raw logs and manifests are stored under reproducibility/. Dataset files are not redistributed in the report; the approved download/preparation scripts and run metadata are included in the repository.

The official Lab 1 PDF requires: independent work for all three tasks; executed notebooks and saved weights; metrics and failure analyses; Task 2 manual review of 20 errors; Task 3 evaluation with the provided script; a 30-sample, two-rater human audit; Kaggle score/rank; and one combined PDF report.

**Submission status:** the results below are evidence-backed. The Task 3 two-rater audit is intentionally marked pending because scores must be collected from two real independent raters; no fabricated ratings are reported.

## 2. Task 1: character-level GPT

Both members trained decoder-only Transformer language models from scratch on TinyStories characters. The models use learned token/position embeddings, causal self-attention, layer normalization, feed-forward blocks, residual connections, and a character vocabulary constructed from training data. No pretrained language model was used.

### Denisha configuration and result

| Item                        | Denisha result                                                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Architecture                | 6-layer decoder-only Transformer; d_model 320; 8 heads; context 256; approximately 3,250,688 parameters                           |
| Training                    | AdamW configuration in task1_llm/member_denisha/config.yaml; 10 epochs; CUDA/Tesla T4 execution                                   |
| Training loss               | 0.7128                                                                                                                            |
| Validation cross-entropy    | 0.6822 nats/character                                                                                                             |
| Validation perplexity / BPC | 1.9783 / 0.9842                                                                                                                   |
| Next-character accuracy     | 0.7822 (78.22%)                                                                                                                   |
| Gradient checks             | gradient_nan_count = 0; final norm 0.2318; max norm 0.2872                                                                        |
| Training time               | 3995.3 seconds; 2.25e5 training tokens/second                                                                                     |
| Evidence                    | task1_llm/member_denisha/outputs/metrics.csv, history.json, samples.json, loss_curves.png, final checkpoint, raw log and manifest |

Denisha's failure analysis covers repetition, spelling instability, and loss of long-range coherence. The model's 256-character context limits story continuity; character-level generation also makes rare-word spelling difficult. Smoke tests passed before training and the notebook generated the required artifacts.

### 3. Task 2: Yelp sentiment classification

Each member trained three models from scratch using learned embeddings: a mean-pooling baseline, a TextCNN, and an experimental GRU model. The official Yelp Polarity test split was evaluated with accuracy, macro/micro/weighted precision, recall and F1, ROC-AUC, PR-AUC, MCC, calibration, confusion matrices, confidence intervals, slice metrics, and McNemar comparisons.

#### Denisha results on the official test split

| Model                | Accuracy | Macro F1 | ROC-AUC | MCC    | Best val loss | Time    |
|----------------------|----------|----------|---------|--------|---------------|---------|
| Baseline mean pool   | 0.9321   | 0.9321   | 0.9804  | 0.8642 | 0.1804        | 26.5 s  |
| Experimental TextCNN | 0.9468   | 0.9468   | 0.9883  | 0.8937 | 0.1376        | 180.3 s |
| Experimental GRU     | 0.9519   | 0.9519   | 0.9901  | 0.9037 | 0.1261        | 159.0 s |

The GRU was Denisha's best model. It achieved 95.19% accuracy, zero gradient NaNs, and the strongest MCC. Its error rates were highest on short reviews, while medium and long reviews were slightly easier. The 20-case error review contains five false positives, five false negatives, five near-threshold cases, and five slice-specific cases, as required.

Evidence: task2\_sentiment/member\_denisha/outputs/metrics\_report.csv, metrics.json, error\_review.json, data\_analysis.json, history files, three checkpoints, results.md, failure\_analysis.md, raw log, and manifest.

## 4. Task 3: CycleGAN image style transfer

The task translates between 300 Monet paintings and 7,038 photographs without paired examples. Denisha trained a CycleGAN with two generators and two PatchGAN discriminators using adversarial, cycle-consistency, and identity losses. Generated images were evaluated with the instructor-provided Part 3 evaluation notebook.

### Denisha official evaluation

| Metric             | Photo -> Monet (B2A) | Monet -> Photo (A2B) | Official combined                          |
|--------------------|----------------------|----------------------|--------------------------------------------|
| FID                | 98.360               | 103.228              | 100.7938                                   |
| MiFID              | 0.4069               | 0.4189               | 0.4129                                     |
| Scored images      | 300                  | 300                  | 300 per direction                          |
| Evaluator device   | CPU                  | CPU                  | Provided evaluation script                 |
| Kaggle team result | -                    | -                    | PairProgramming_Team_09: -50.6033, rank 34 |

Additional Denisha metrics include directional FID, KID, generative precision/recall, LPIPS evaluation records, cycle histories, loss curves, generated image archives, and the official two-row submission.csv. The Kaggle submission was produced from the trained CycleGAN outputs and was not manually edited.

Failure analysis identified poor handling of dark scenes, texture and brush-stroke artifacts, washed-out colors, and content loss in foggy or low-contrast images. The model's training checkpoint and epoch checkpoints are stored in the artifacts; the approximately 340 MB latest checkpoint should be stored in Google Drive rather than normal GitHub storage.

### Human audit requirement

The Lab requires two independent raters to score 30 fixed samples for style, content, and artifacts, followed by inter-rater agreement. The repository currently contains the 30-row audit template but no ratings. This section must be updated after both raters independently inspect the images; scores must not be invented or copied.

## 5. Cross-task comparison and limitations

Denisha's Task 1 run provides a compact from-scratch character model with stable gradients and a validation BPC of 0.9842. Her Task 2 GRU is the strongest of her three classifiers at 95.19% accuracy and 0.9037 MCC. Her Task 3 run produced a valid official submission and a combined official FID of 100.7938, while the public Kaggle result for PairProgramming\_Team\_09 was -50.6033 at rank 34.

The main limitations are single-run training for each model, different hardware between members, the small Monet domain, variance in FID from only 300 scored images, and the remaining human-audit evidence. Results should therefore be interpreted as reproducible single-run experiments rather than estimates of uncertainty over random seeds.

Recommended next steps are to complete the two-rater audit, record Cohen's kappa or percent agreement, attach the large checkpoint through Google Drive, execute/save any notebook versions required by the instructor, and package the final ZIP with Parts 1–3 and this report.

## 6. Evidence map

| Requirement                             | Denisha evidence                                                                                                                                                                   |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 1 notebook, weights, outputs       | task1_llm/member_denisha/Task1_GPT_Colab.ipynb; checkpoints/gpt_from_scratch.pt; outputs/metrics.csv, history.json, samples.json, loss_curves.png; results.md; failure_analysis.md |
| Task 2 notebook, three weights, metrics | task2_sentiment/member_denisha/Task2_Sentiment_Colab.ipynb; three checkpoint files; outputs/metrics_report.csv, metrics.json, error_review.json; results.md; failure_analysis.md   |
| Task 3 code, weights, outputs           | task3_gan/member_denisha/Task3_CycleGAN_Colab.ipynb; run_task3.py; checkpoint archive; outputs/images.zip; official metrics and submission.csv                                     |
| Reproducibility                         | reproducibility/raw_logs/ and reproducibility/manifests/                                                                                                                           |
| Shared repository                       | <a href="https://github.com/Shibin-1020/Gen-AI---LAB-1">https://github.com/Shibin-1020/Gen-AI---LAB-1</a>                                                                          |
| Pending before final ZIP                | Two-rater human audit, agreement statistic, final Drive checkpoint link, and final packaging validation                                                                            |

References: DATA266 Lab 1 official handout; Vaswani et al., Attention Is All You Need; Zhu et al., Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks; the instructor-provided Part 3 evaluation script; official Yelp Polarity and TinyStories data sources.